

A Review of Dynamic Operating Envelopes: Computation, Applications and Challenges

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Abstract: The integration of Distributed Energy Resources (DERs) into power grids presents significant challenges to grid performance, requiring innovative solutions for effective operation. Dynamic Operating Envelopes (DOEs) offer a promising approach by optimizing the use of existing infrastructure while ensuring compliance with network constraints. This paper reviews various DOE calculation methodologies, focusing on Optimal Power Flow (OPF)-based methods. Key findings include the role of DOEs in optimizing import and export limits, with critical factors such as forecast accuracy, network modelling, and the effects of mutual phase coupling in unbalanced networks identified as influencing DOE performance. The paper also explores the integration of DOEs into smart grid frameworks, examining both centralized and decentralized control strategies, as well as their potential for providing ancillary services. Challenges in scaling DOEs are also discussed, particularly regarding the need for accurate forecasts, computational resources, communication infrastructure, and balancing efficiency and fairness in resource allocation. Lastly, future research directions are proposed, focusing on the practical application of DOEs to improve grid performance and support network operations, as well as the development of more robust DOE calculation methodologies. This review provides a comprehensive overview of current DOE research and identifies avenues for further exploration and advancement.

Keywords: dynamic operating envelopes; optimal power flow; uncertainties; distributed energy resources; forecasting accuracy; power grid stability



Academic Editor: Wei Gao

Received: 4 March 2025

Revised: 25 March 2025

Accepted: 31 March 2025

Published: 3 April 2025

Citation: Wickramasinghe, A.; Vilathgamuwa, M.; Nourbakhsh, G.; Corry, P. A Review of Dynamic Operating Envelopes: Computation, Applications and Challenges. *Modelling* **2025**, *6*, 29. <https://doi.org/10.3390/modelling6020029>

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1. Introduction

Renewable energy is gaining more interest today due to its clean, inexhaustible nature and growing competitiveness. Among different renewable energy sources, solar power is currently the fastest-growing, surpassing others like wind, hydro, and bioenergy, particularly in Australia. However, the rapid integration of Distributed Energy Resources (DER)—such as distributed solar Photovoltaic (PV) systems, Battery Energy Storage Systems (BESS), Electric Vehicles (EVs), and other controllable loads—has introduced significant challenges in managing bidirectional power flows within the electrical grid. To mitigate these challenges, Distribution Service Operators (DSO) have implemented static limits (kW/customer) to prevent over-voltage and reverse power flows, considering ‘worst case scenarios’ like minimum power demand, maximum local generation, and weak networks [1]. For instance, one of the DSOs in Queensland (Energex) imposes a 5 kW export limit for residential customer connections. As more customers install DERs on

their premises and until battery systems become economically viable, these static limits are expected to become stricter in the future.

Different solutions have been proposed to maximize DER integration while ensuring the safe and optimal operation of the network. Ref. [2] proposed the use of On-Load Tap Changers (OLTC) to mitigate overvoltage; however, these devices suffer from slow response rates and mechanical wear. A volt/var and volt/watt approach is used in [3] to maximize the PV hosting capacity, but achieving compliance with standards governed by Distribution Network Service Providers (DNSP) remains a challenge [4]. Integrating BESS at customer premises can address these issues, but the high capital costs remain a significant barrier [5]. Custom power devices can dynamically regulate voltages by adjusting between capacitive and inductive modes to mitigate over-voltage, but their complex control systems limit their responsiveness to sudden voltage surges [6].

DOEs offer a promising alternative, providing flexible import and export limits at each DER connection or customer connection point. These limits adapt in real-time, based on factors such as time of day, location, and network conditions. DOEs ensure power flows remain within safe limits while maximizing the utilization of existing network assets by adjusting active and reactive power flows, ensuring power flows remain within safe limits while maximizing the utilization of existing network assets. By dynamically adjusting active and reactive power flows, DOEs facilitate efficient DER integration without overloading network infrastructure, thereby enhancing grid stability and operational flexibility. However, accurate demand and generation forecasts, as well as full network visibility or historical smart meter data, are essential for effective DOE implementation [1].

The advantage of DOEs over the aforementioned methods is their lower capital costs, as they do not require expensive infrastructure investments. Additionally, DOEs enable maximum DER integration by dynamically adapting to real-time conditions and specifying the available capacity for power imports and exports during predefined time intervals. This ensures the physical and operational constraints of the network are not violated [7]. Furthermore, DOEs help DSOs optimize the use of existing infrastructure, such as poles and wires, reducing the need for costly network upgrades. Figure 1 shows how dynamic exports and imports behave in comparison to static limits.

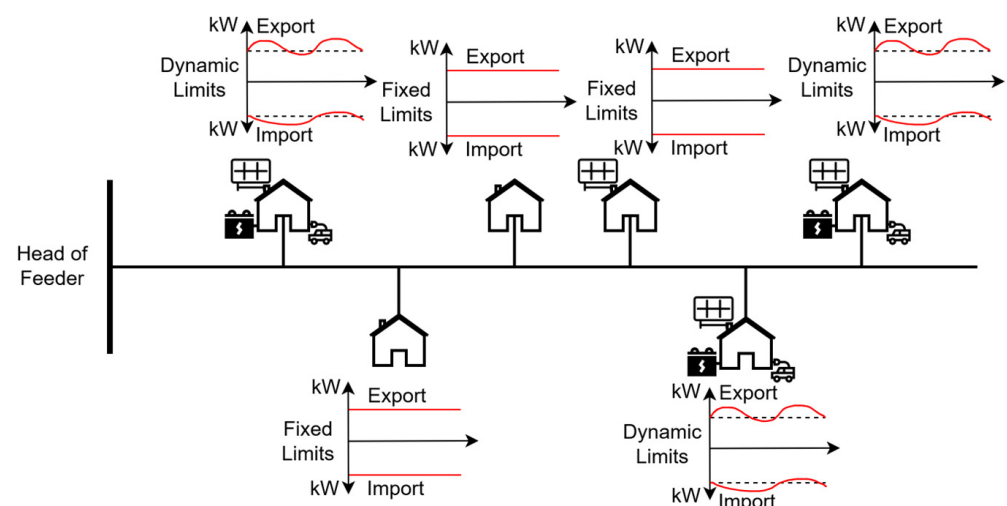


Figure 1. Fixed limits and Dynamic Limits.

Over the past few years, DOEs have gained momentum as a viable approach for providing prosumers with both dynamic import and export limits and have advanced to practical implementation in Australia. Recent literature has increasingly focused on their derivation, application, and viability [8–11]. Notably, in Australia, pilot projects like Project

Energy Demand and Generation Exchange (EDGE) [10] and Project Energy Value Optimization Leveraging Envelopes (EVOLVE) [11] have demonstrated the practical implementation of DOEs, yielding promising results. These projects, funded by the Australian Renewable Energy Agency (ARENA) [12], have been instrumental in advancing the understanding of DOEs.

The main contributions of this review are to highlight the significance of Dynamic Operating Envelopes (DOEs) in proactively managing network constraints to ensure the safe operation of electricity networks. It examines the current frameworks proposed for integrating DOEs into network operations, with a particular focus on leveraging DOEs to enhance coordination between distributed energy resources (DERs) and the grid. Furthermore, the paper explores various DOE calculation methodologies, emphasizing optimization-based approaches using AC Optimal Power Flow (ACOPF) due to its capability to model power flow constraints accurately. Additionally, key challenges associated with DOE calculations are discussed, along with potential avenues for future research, including the exploration of DOE data to assess end-user flexibility for providing additional grid services.

The remainder of this paper is organized as follows: The rest of Section 1 discusses the key stakeholders in DOE implementation, dynamic limits for imports and exports, and the calculation frequency required to align with market dispatch intervals. Section 2 reviews different methodologies for calculating DOEs, comparing their strengths and limitations. Section 3 investigates the impacts of using ACOPF relaxations for DOE calculations. Section 4 explores the importance of selecting an objective function that strikes a balance between fairness and maximization in DOE calculations, as demonstrated through a case study. Sections 5 and 6 identify key challenges and outline future research directions aimed at enhancing DOE robustness by leveraging DOEs for broader grid services, respectively.

1.1. Stakeholders Associated with Dynamic Operating Envelopes

For centralized DOE calculations, a central authority must be responsible for calculating the DOEs. This authority should have access to accurate forecasts, which necessitate access to smart meter data, as well as a reliable and up-to-date network model. Additionally, the central authority must possess the computational and communication capabilities to perform near real-time operational DOE calculations and effectively share them with each consumer's inverter system/s. The proposed hybrid framework for DOE integration in Australia, along with the associated entities, is illustrated in Figure 2 [13]. In this hybrid framework, the DSO would oversee and communicate distribution network constraints (DOE) to each DER either through aggregators or retailers. Additionally, these DOEs are shared with the Australian Energy Market Operator (AEMO), who is proposed to be responsible for optimizing DER bids for both wholesale electricity markets and system support services. While Static Operating Envelopes (SOEs)/fixed limits are set in advance, DOEs are calculated by the DSO either a day ahead or in real-time.

1.2. Maximum and Minimum Dynamic Limits

While DOEs allow for variable imports and exports, it is essential to set both upper and lower limits for these values. When calculating the DOE using an Optimal Power Flow (OPF)-based approach, the optimization problem must include constraints on these limits to ensure a feasible solution. Typically, the export limit is set between 1.5 kW and 10 kW [14], although this range may vary depending on the specific DSO. In cases where the internet connection between the consumer's DER inverter and the DSO fails, the export capacity is reduced to the minimum limit, ensuring that the system operates within the specified boundaries despite the communication disruption. Currently, there is no defined

cap on the import limit set by the DSO, but in the EDGE project [15], an import limit of 14 kW was considered based on a 60 A short circuit fuse rating.

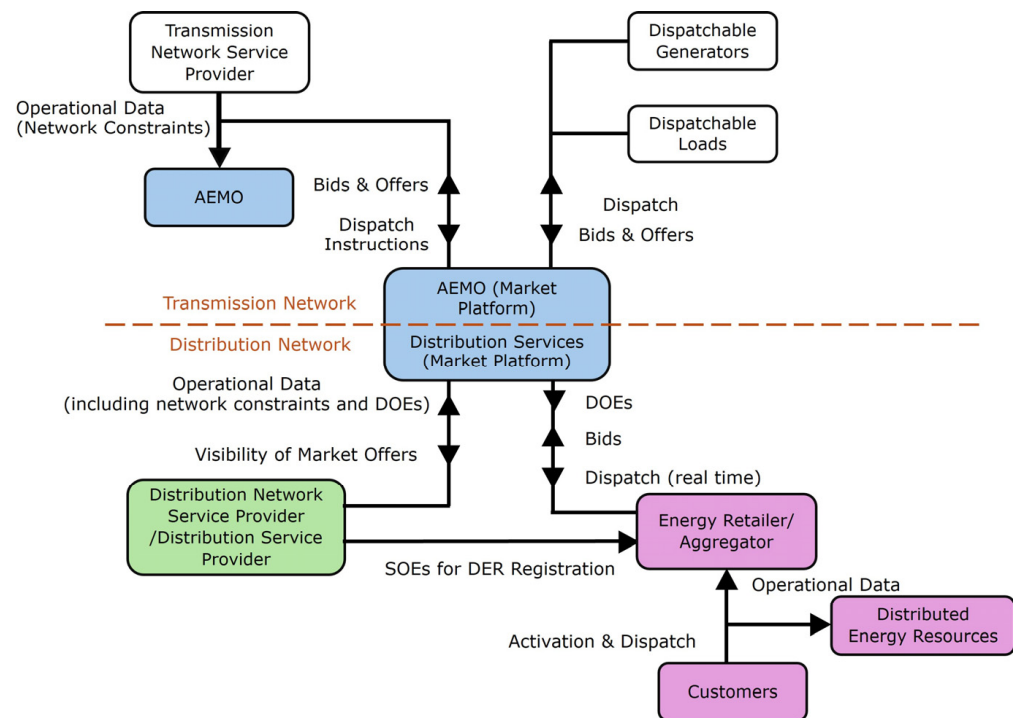


Figure 2. Hybrid framework developed in the open energy networks project [13].

1.3. Temporal Resolution for Dynamic Operating Envelopes

The calculation frequency of DOEs is influenced by household load, generation patterns, and their forecasts. It is essential that the time resolution of DOEs matches the market dispatch interval, allowing aggregators to engage in wholesale electricity markets while maintaining network security. Ref. [1] also highlights the importance of aligning DOE calculations with the National Electricity Market (NEM) dispatch interval [16], recommending updates every 5 min.

2. DOE Calculation Methodologies

There are various algorithmic methods to calculate an operating envelope mathematically. These methods differ from each other based on the model of the distribution network and the connected DER at each node. Irrespective of the specific calculation techniques used to obtain DOEs, they must accurately reflect electrical engineering principles while accounting for the physical and operational network limits, including voltage and thermal constraints, to ensure a valid operating envelope. Regardless of the chosen network model, mathematical model, and algorithm, the necessary steps to compute an operating envelope for each time interval are illustrated in Figure 3 as suggested in the knowledge sharing report on DOE published by the Australian National University (ANU) [7].

Based on the literature and the above-mentioned pilot projects, DOE calculation methodologies can be divided into several criteria as shown in Figure 4. These methods are centralized and DOEs are calculated by the DSO. Model-aware methods require full network visibility, which some DSOs may not have [17]. On the other hand, model-free methods rely on customer smart meter data to calculate the DOE, but as not all customers have smart meters installed in their premises, this could pose challenges such as data gaps, reduced accuracy, and the need for alternative estimation methods such as Distribution System State Estimation (DSSE) [18,19]. So, each method presents different challenges.

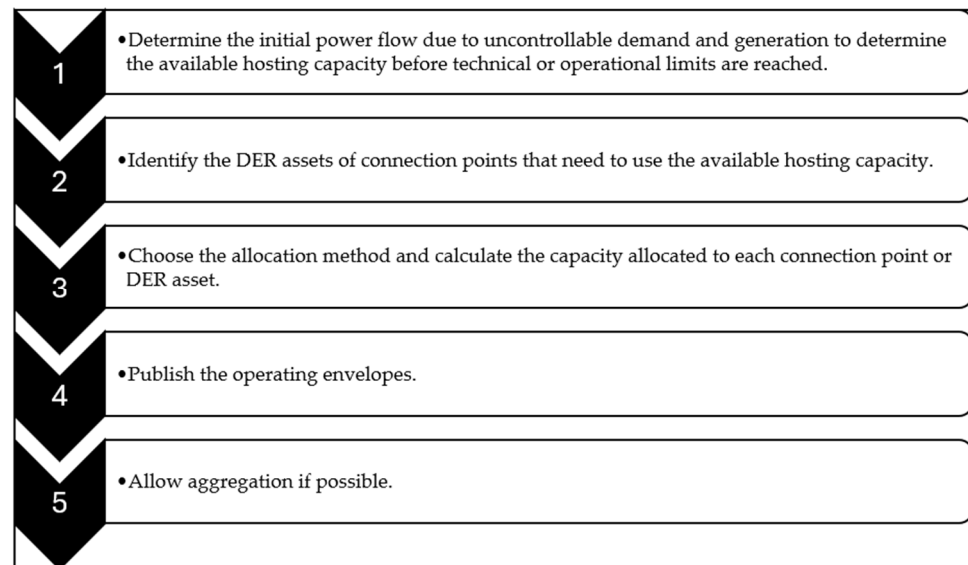


Figure 3. Necessary steps to calculate DOE as suggested in [7].

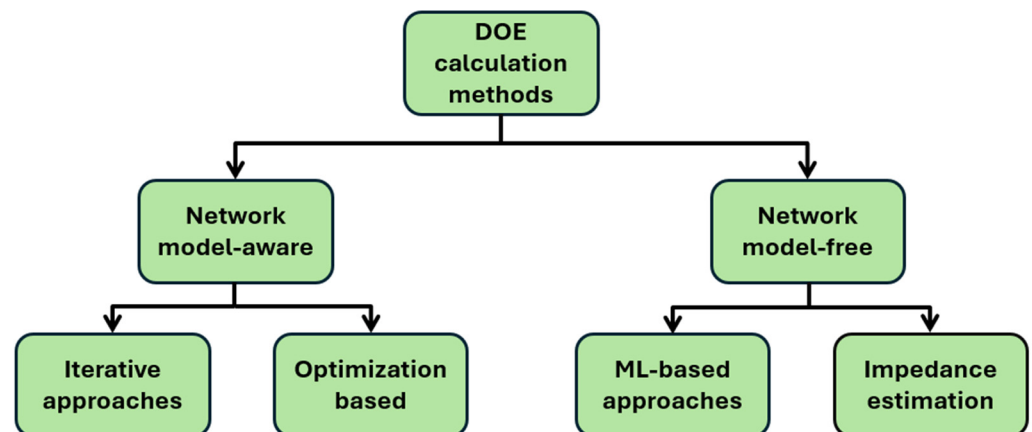


Figure 4. Different DOE calculation methodologies.

2.1. Model-Aware Dynamic Operating Envelopes

2.1.1. Iterative Approaches

An iterative approach was first utilized for calculating DOE in [10], where the maximum allowable imports and exports are allocated, and load flow is executed for each time step. If any of the network constraints (current and voltage) are violated, the allocated limit is deducted iteratively until all constraints are satisfied. In [20], a bi-section method was used to allocate export values for consumers. Using iterative approaches can be slow and can take a long time to converge; even so, it might not be the optimal solution.

2.1.2. Optimization Based Approaches

Optimization based approaches for calculating DOEs generally follow a structured methodology as shown in Figure 5. The parameters for the optimization problem are typically presented as inputs, while the calculated import and export limits serve as the outputs.

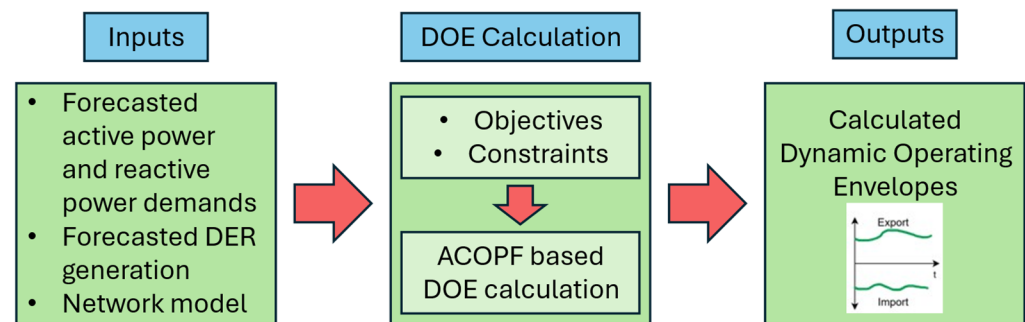


Figure 5. DOE calculation using optimization-based approaches.

1. Objective function

For any optimization problem, the objective function plays a crucial role as it determines the goal of the optimization and directly influences the solution. This holds true when using an optimization-based approach to calculate DOEs. The calculated DOEs can vary significantly depending on the chosen objective function. Based on the literature, DOE calculation objective functions can be categorized into two main categories:

- Maximize DOE

This approach focuses on maximizing the total sum of DER exports/imports for all customers as much as possible without violating system constraints. Although this method results in a higher total sum of exports and imports, it disproportionately disadvantages customers at the end of the feeder, as their import and export capacities are limited due to voltage constraints and network impedance. In contrast, customers closer to the distribution transformer benefit more, as their total network impedances are lower and they therefore gain access to a larger share of the available network capacity. This imbalance can lead to an inequitable distribution of operating envelopes, raising concerns about the fairness of the participating prosumers. Figure 6 illustrates the export envelopes calculated for different customers using the objective function that maximizes total exports. The blue-colored operating envelope represents the consumer at the far end of the feeder, whose export capacity drops to zero during solar peak hours. This highlights the unfairness faced by consumers with higher impedances, as they are unable to fully utilize their export potential.

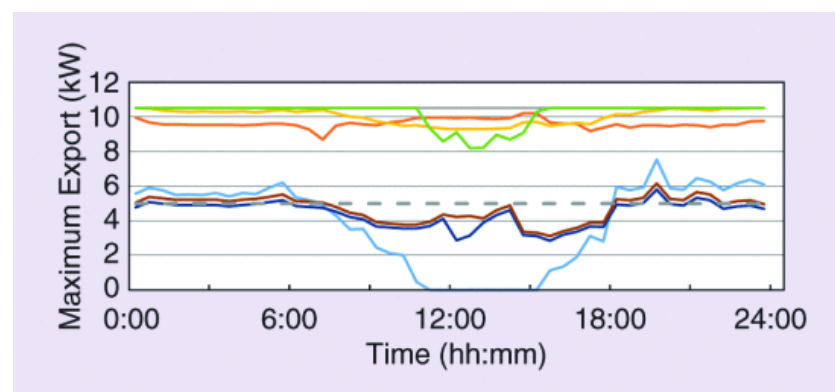


Figure 6. Maximum allocation of DOE as per [9].

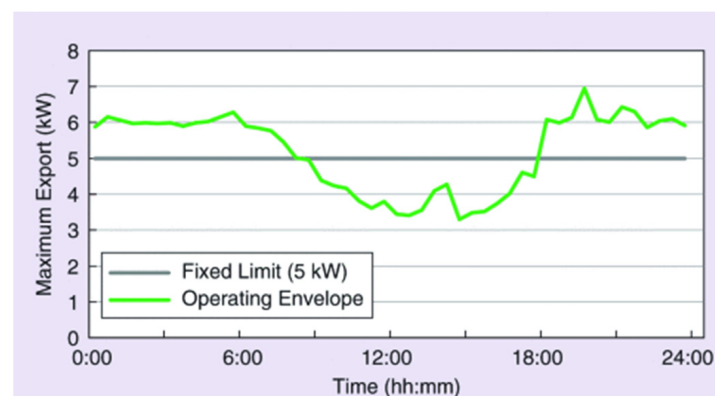
- Fair Allocation Approaches

This approach ensures equitable division of network capacity among all participants, preventing monopolization of network resources. For achieving fair operating envelopes, different objective functions can be formulated. Some of these, as found in the literature, are shown in Table 1.

Table 1. Formulations of different objective functions for fair DOE allocation.

References	Objective Function
[21,22]	Maximise the smallest DOE
[23]	Minimize the squared difference between the calculated DOE and the forecasted PV generation
[8]	Maximize the multiplication between the customer inverter capacity and logarithmic value of the calculated DOE

Though these objective functions are different from each other; they all try to incorporate fairness into the objective function, but by doing so can reduce the sum of total imports and exports of all the customers. Figure 7 shows the calculated export envelopes when the same envelope is allocated to all customers in the feeder. Although this method limits the export capacity of some consumers, it ensures a fairer distribution of network capacity, providing equal access to all participants regardless of their location on the feeder.

**Figure 7.** Allocation of the same operating envelopes for a fair approach [9].

2. Constraints

In the optimization problem, the main constraints should satisfy the network limits and power balance equations. The primary constraints that must be met include:

- Voltage constraints

Within the distribution network, voltage limits are predefined for all nodes at a specific voltage level. These limits vary across different segments of the electricity distribution network, with each segment having a nominal voltage along with upper and lower voltage boundaries. The Australian standards AS600038-2012 [24] for voltage in low-voltage distribution networks are provided in Table 2. Hence the voltage constraint should be fulfilled as $V_{min} \leq |V_i| \leq V_{max}$, where V_i is the voltage at node i and should be within 216–253 V. It is important to note that the voltage at a given node in the network is directly influenced by power imports and exports at other nodes, as they share a common electrical path. Consequently, both upstream and downstream nodes can impact the voltage behavior at the selected node.

Table 2. Voltage standards as per AS600038-2012.

Australian LV Standard	Voltage
Nominal Voltage	230 V
Upper Limit (+10%)	253 V
Lower Limit (−6%)	216 V

- Thermal constraints

Thermal limits define the maximum apparent power that can flow through a branch in the electricity distribution network. These limits depend on various factors, including the type of asset (such as conductors or transformers) and specific asset characteristics (such as conductor size). In a distribution network, both real and reactive power components contribute to the total apparent power in a branch. The constraint that should be satisfied is $P_{ij} + Q_{ij} \leq (S_{ij}^{max})^2$. Where, P_{ij} , Q_{ij} and S_{ij}^{max} are the active power, reactive power, and maximum thermal limit, respectively.

- Line constraints

Line constraints represent the maximum amount of current (ampacity) that a conductor can carry without exceeding its temperature limit.

- Active Power Balance and reactive power balance

Based on the selected network model (Branch flow model or Bus injection model [25]), this constraint varies. The Branch Flow Model represents power flows along distribution lines, enforcing constraints directly on branch currents, voltages, and power losses [26]. In contrast, the Bus Injection Model formulates constraints at the node level, using power injections at buses while considering network admittance relationships [26]. Though each model has a different mathematical formulation, both ensure that power flow constraints are met while optimizing network operations, and the branch flow model is usually used in radial structures.

- Voltage unbalance constraint

Ref. [27] analyzed the impact of incorporating the voltage unbalance constraint in DOE calculations using the Voltage Unbalance Factor (VUF) metric and found that neglecting this constraint can lead to overly optimistic and inaccurate import/export limits.

2.2. Model Free Dynamic Operating Envelopes

Since distribution companies, particularly in Australia, often lack complete electrical models needed to calculate correct individual impedance or admittance values for each customer [17]; using incorrect impedance data can result in suboptimal DOEs that may not guarantee network constraints. To address this issue, model-free DOE calculations have been introduced. These can be achieved either by estimating impedance values using smart meter data—such as active power, reactive power, and voltage measurements—or by leveraging Machine Learning (ML) approaches.

2.2.1. Machine-Learning Based Approaches

DOEs can be calculated without network models using Neural Networks by leveraging smart meter data including active power, reactive power, and voltage measurements from the customers [28]. This method is faster than load flow-based techniques as it relies on model-free voltage calculations. The proposed method consists of three main parts.

As shown in Figure 8, the first step involves pre-processing the smart meter data, which includes customer active power, reactive power, and voltage measurements. The data are then filtered to remove abnormal values and near-zero values, ensuring a cleaner and more reliable dataset. The second step is the offline calculation phase, where multiple Neural Networks (NNs) are trained using pre-processed smart meter data. These trained NNs are then evaluated using a testing dataset, and based on the testing results, the best-performing NN is selected for use in the online phase. In the third step, the selected NN is applied online in combination with a heuristic algorithm to validate different import and export envelopes for active customers. The NN also evaluates voltage, thermal, and

transformer constraints while calculating model-free voltages. This method can be applied even with a smart meter coverage as low as 20% [29]. The main drawback of this approach is the need for historical smart meter data from a significant number of customers to effectively train the NN. Additionally, the NN is not universal and needs to be retrained for different network models.

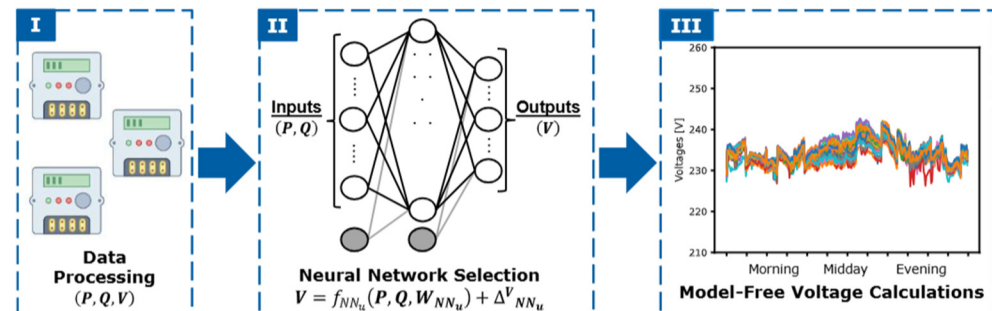


Figure 8. Model-free DOE calculation approach according to [29].

2.2.2. Impedance Estimation Approaches

In [30], a novel method was developed to estimate impedance for each customer using linearized equations derived from an unbalanced three-phase, four-wire model. The approach estimates impedance for each consumer by minimizing the loss function between the measured smart meter data and the calculated active power, reactive power, and voltage values. While this method is valuable in cases of partial or no network visibility, it assumes full smart meter coverage and requires prior knowledge of each customer's phase connection. Additionally, the linearization process introduces errors in the estimated impedance values.

3. Utilization of Alternating Current (AC) OPF for DOE Calculation

Optimal Power Flow, which was first introduced by Carpentier in 1962 [31], is a critical tool in power system operation and planning. Any optimization problem aimed at enhancing the operation of an electric power system while adhering to the physical laws governing electrical engineering and physical network constraints falls under the domain of OPF. It determines the optimal operating conditions for power systems while satisfying system constraints and minimizing or maximizing a specific objective, such as cost, losses, generation, or emissions. In a classic economic dispatch problem, the decision variables are power outputs of each generator, while the constraints include power flow equations (for balancing power and demand), maximum generator capacity, and network constraints (voltage, thermal, and current). The objective is a cost function of generation, usually to determine the power generated by each generator unit in a way that minimizes the total operational cost [32]. OPF problems are foundational in power system studies and have been widely researched for decades. OPF is frequently utilized for calculating DOE according to [8,23,33], where the objective function is modified to maximize DER generation (for exports) within network constraints. However, when dealing with ACOPF, the problem becomes non-convex and non-linear, primarily introduced by power flow equations. These equations involve trigonometric functions that describe the relationship between voltage, current, and power, leading to scalability issues and a computationally intensive optimization process. The non-convexity introduces multiple local minima, which makes finding the global optimal solution difficult, especially for large-scale power systems. Hence many authors have used different relaxations for converting these nonlinear equations to linear or convex. These relaxations aim to simplify the problem and make it more computationally tractable. Figure 9 illustrates the difference caused by convex relaxations and approximations in the original non-convex feasible region of ACOPF.

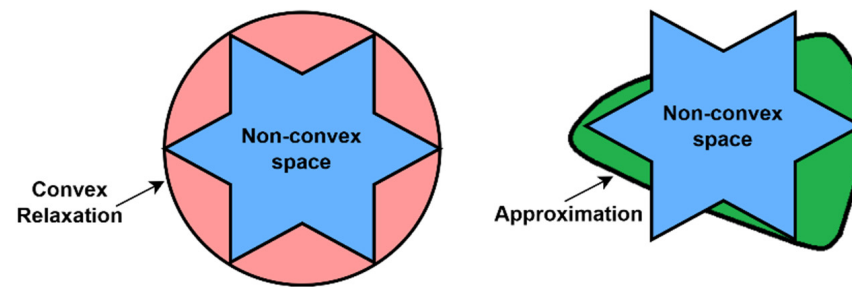


Figure 9. Convex relaxation and approximation.

Impact of Using Relaxations for DOE Calculation

As ACOPF optimization problems include power flow equations as constraints, they are usually NP-Hard [34] and can converge to a suboptimal solution even for radial network models. Although relaxations or approximations can simplify the problem, they often do not fully satisfy the non-linear power flow equations. The goal of these methods is to create more manageable formulations that still provide an adequate representation of the actual physical behavior of the power system. Convex relaxations offer valuable properties such as providing bounds on the optimal objective for the original non-convex problem and offering sufficient conditions for verifying whether the problem is infeasible. Some convex relaxations can even guarantee global optimality for specific classes of problems, with certain conditions that can be evaluated either before or after solving the relaxation. Even though some of the relaxations used in OPF formulations proved to be exact for some scenarios [35], it might not be the same for some applications. In [21], it is shown that employing a Second-Order Cone Programming relaxation (SOCP) for DistFlow equations can indeed result in inaccurate DOEs.

4. Case Study

In the considered case study customer demand is considered uncontrollable. Hence only export limits are considered. The export envelopes are implemented on a real LV residential network in Australia, as shown in Figure 10 [36]. The residential customer demand profiles, sampled at 5 min intervals, are from a summer weekday in Australia obtained from anonymized smart meters [14]. Based on the most popular monthly solar installations in Australia since 2022 [37], inverter capacities of 5 kW, 6 kW, 8 kW, and 10 kW are chosen and assigned to each node.

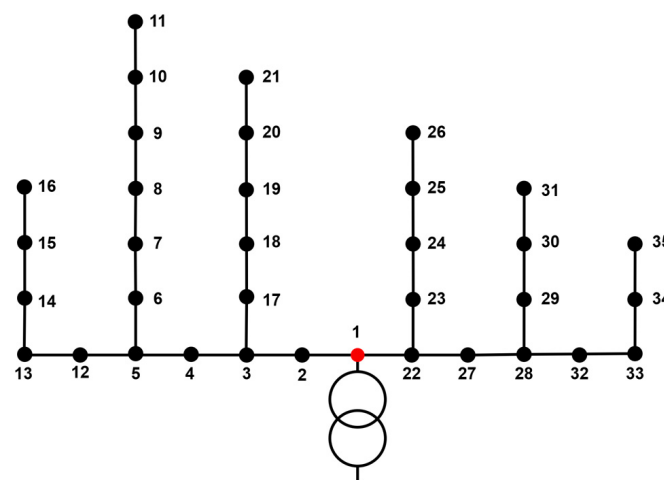


Figure 10. Considered LV network of 34 nodes (excluding slack bus, shown in red).

4.1. Network Model

The paper follows the graph theory notation, where $\mathcal{G}$ is an undirected graph such that $\mathcal{G} = (\mathcal{N}, \mathcal{L})$ with $\mathcal{N}$ representing the set of nodes (buses) and $\mathcal{L}$ representing the set of edges (lines) in the distribution network. Since $\mathcal{G}$ is undirected $(i, j) \in \mathcal{L}$ and $(j, i) \in \mathcal{L}$. For simplicity, we assume the network is balanced and can thus be represented by its single-phase equivalent circuit as shown in Figure 10. All consumers are considered to have opted in for DOE. Consider a LV distribution network with a set of nodes $n \in N$, where $n \in \{1, 2, \dots, N\}$. These nodes are connected through a set of distribution lines $\ell \in \mathcal{L}$ where $\ell \in \{1, 2, \dots, \mathcal{L}\}$. Consumers, indexed by $i \in I$, where $i \in \{1, 2, \dots, I\}$, are each connected to a different node in the network. Time intervals are represented by a set of $t \in T$, where $t \in \{1, 2, \dots, T\}$. The net active power exports for each prosumer ($P_{i,t}$) at time interval t is,

$$P_{i,t} = p_{i,t}^{DOE} + p_{i,t}^{PV} - p_{i,t}^{dem} \quad \forall i \in I, \forall t \in T \quad (1)$$

where $p_{i,t}^{DOE}$ is the active power export limits for each customer, $p_{i,t}^{PV}$ is the PV generation from each customer and $p_{i,t}^{dem}$ is the total active power demand by uncontrollable loads for each prosumer. As the focus here is to calculate active power operating envelopes considering a unity power factor, the net reactive power for each prosumer ($Q_{i,t}$) at time interval t is,

$$Q_{i,t} = -q_{i,t}^{dem} \quad \forall i \in I, \forall t \in T \quad (2)$$

where $q_{i,t}^{dem}$ is the total reactive power demand by uncontrollable loads for each prosumer.

4.2. Formulation of OPF

The export envelopes will be calculated under three different objective functions to examine the impact of fairness-based and export-maximization-based approaches. This comparison allows for an evaluation of the trade-offs between maximizing energy export and ensuring a fair distribution of benefits among participants. The fairness-based objective ensures that the energy export is equitably shared, preventing any single participant from dominating, which is especially important in systems with multiple prosumers. On the other hand, the export-maximization-based objective focuses on optimizing the total energy export, maximizing economic or environmental benefits, particularly in systems aiming to enhance renewable energy integration. By testing both objective functions, the study aims to assess which approach best aligns with the involved stakeholder's goals and the overall system objectives, providing a balanced solution that addresses both equity and efficiency.

4.2.1. Objective Functions

Objective 1—Maximise Total Exports

$$\max \sum_{i=1}^N p_{i,t}^{DOE} \quad \forall i \in I, \forall t \in T \quad (3a)$$

Objective 2—Maximise the smallest DOE

$$\max \sum_{i=1}^N \bar{p}_{i,t}^{DOE} \quad \forall i \in I, \forall t \in T \quad (3b)$$

where $\bar{p}_{i,t}^{DOE}$ is the smallest export limit.

Objective 3—Minimize the squared difference between the calculated DOE and the forecasted PV generation

$$\max \sum_{i=1}^N (p_{i,t}^{DOE} - p_{i,t}^{PV})^2 \forall i \in I, \forall t \in T \quad (3c)$$

4.2.2. Constraints

The constraints of the formulated optimization problem are as follows:

$$0 \leq p_{i,t}^{DOE} \leq p_{i,t}^{PV} \forall i \in I, \forall t \in T \quad (4)$$

$$V_{i,t} - V_{j,t} = z_{\ell} I_{\ell,t} \forall i \in I, \forall \ell \in \mathcal{L} \quad (5)$$

$$\sum_{k:k \rightarrow i} I_{ki} - \sum_{j:i \rightarrow j} I_{ij} = \sum_{i=1}^I \frac{P_{i,t} - jQ_{i,t}}{(V_{i,t})^*} \forall n \in N, \forall i \in I, \forall t \in T \quad (6)$$

$$V_{min} \leq |V_{i,t}| \leq V_{max} \forall i \in I, \forall t \in T \quad (7)$$

$$|I_{\ell,t}| \leq I_{max} \forall \ell \in \mathcal{L}, \forall t \in T \quad (8)$$

Equation (4) limits the calculated export limit for each prosumer so that it does not exceed the forecasted PV generation. The voltage drop along the line, as described in constraint (5), relates to the node voltages between two consumers at time t , where z_{ℓ} is the impedance between the two, and $I_{\ell,t}$ is the current flowing along the distribution line at t . Using Kirchoff's current law in (6), where I_{ki} is current flowing into line ki from node k to node i , and I_{ij} is current flowing out of line ij from node i to j , adjusted by the current demand at node i . Equations (7) and (8) represent the voltage and current limits, respectively.

4.2.3. Comparison of Results

The calculated export intervals are compared in Table 3. The first column shows the different objective functions, while the second column presents the calculated export limits for each prosumer. The export envelopes for the consumer at the end of the feeder (node 35 in Figure 9) are shown in red, highlighting the variation in export limits based on each objective function. The third column displays the voltage profiles corresponding to the calculated export envelopes. The last column is the summation of total exports, showing that the largest export value is provided by objective (3a). Additionally, objective (3c) incorporates fairness while achieving exports close to the maximum. In contrast, the (3b) objective function yields the lowest export envelopes. Therefore, it can be concluded that objective (3c) strikes a balance between maximizing exports and ensuring fairness.

Table 3. Comparison of export envelopes calculated using different objective functions.

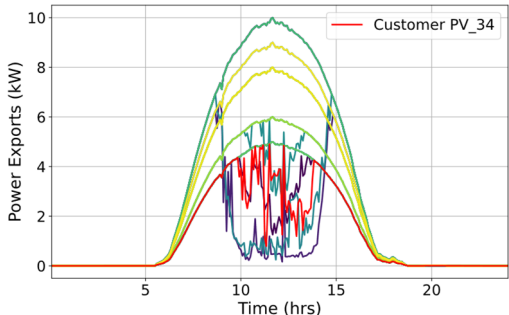
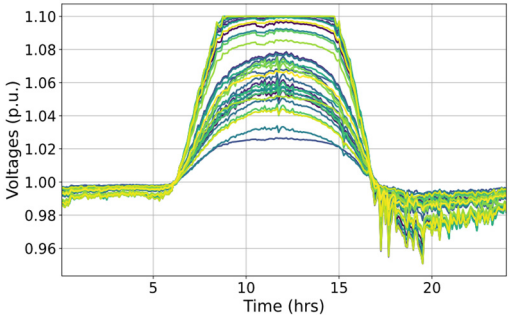
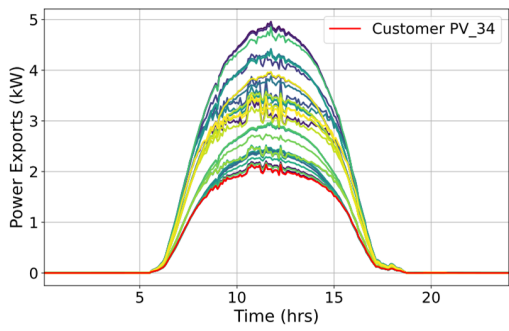
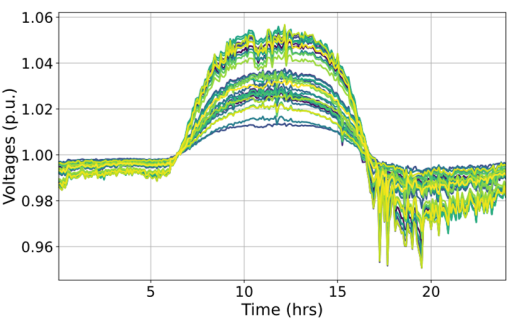
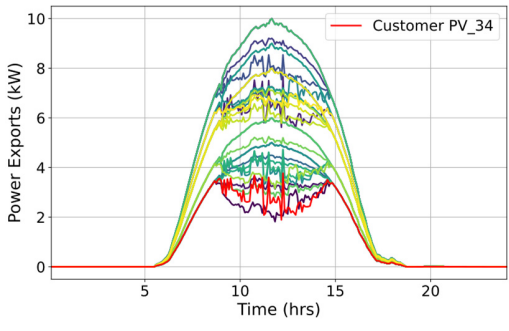
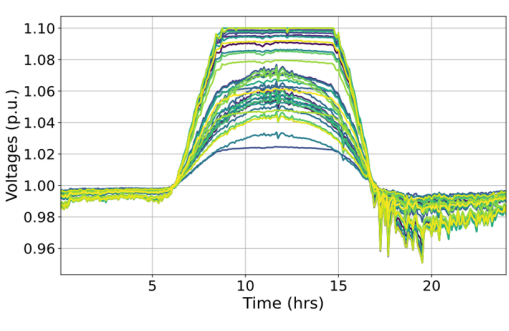
Objective Function	Export Envelopes	Voltage Limits	Total Exports (MW)
(3a)			20.398

Table 3. Cont.

Objective Function	Export Envelopes	Voltage Limits	Total Exports (MW)
(3b)			10.091
(3c)			19.970

5. Issues Associated with DOE Calculations

This section examines the technical challenges associated with the calculation and implementation of DOEs.

5.1. Centralized DOE Calculations

In Australia, active prosumers are currently enrolled in these DOE schemes as they are offered by major network distribution companies such as Energex [14] and AusNet [38]. Calculating DOEs is presently carried out by DSO in a centralized manner. Once the DOEs are calculated, they are broadcasted to the smart inverters of prosumers for a predefined time interval, and the import/export limits at either each customer's DER connection or at each customer's connection are adjusted as per the published DOE limits [7]. With more and more customers enrolling in DOE programs, calculating DOE for each customer individually will be a challenge for DSO operators, as the valid time interval for each DOE is limited to 5–30 min [7]. So, within the specified time interval, DSO must calculate and communicate DOE limits for every participating customer. Centralized DOE calculations by DSO can lead to several challenges, such as scalability issues, privacy concerns, and computational bottlenecks. Traditional centralized methods can struggle to accommodate DERs growing complexity and variability, especially with the high penetration of renewable energy. Decentralized methods facilitate decision-making at a localized level, improving the efficiency of resource use within the network while preserving overall system stability. In contrast, distributed methods offer enhanced scalability and flexibility, ensuring compliance with operational constraints even in highly variable conditions. Hence, exploring novel ways to calculate distributed DOEs can be vital as more and more customers adopt flexible exports.

5.2. Accurate Forecasts

There is a degree of consistency in the forecasting period provided by DSOs that currently offer DOEs to their customers. Typically, these forecasts cover the next 24 h, with data provided at 5 min intervals [39]. A study in [40] examined the effect of using 30 min ahead, 6 h ahead (intra-day), and 24 h ahead (day-ahead) forecasts for active power and voltage on DOE calculation, finding that using a day-ahead forecast lead to significant constraint violations in the calculated DOE. Therefore, it is important to note that the shorter the forecast period relative to the actual DOE calculation interval, the more accurate the resulting operating envelopes will be. Additionally, Ref. [41] explored the impact of forecast errors on DOE calculations. Moreover, for accurate forecasts, DSOs need access to smart meter data, but only 10–50% of customers currently have smart meters installed. Due to the regulatory separation between network service provision and retail, DSOs must purchase this data from retailers, but only for the previous day, which adds further complexity [22]. Though there are different forecast methods for DER and consumer demand [42–44], they are inherently imperfect and not entirely precise. These forecasts rely on historical data and predictive models that may not fully account for abrupt changes in consumption patterns, weather variations, or fluctuations in DER output. This uncertainty introduces discrepancies between predicted and actual values, which can result in deviations from optimal DOEs and lead to potential violations of grid constraints. Although forecasting accuracy has improved with technological advancements, complete precision in predicting demand and DER behavior is yet to be explored. Furthermore, Model Predictive Control (MPC) [45]-based methods can be utilized to address these uncertainties by dynamically adjusting operational decisions based on updated forecasts and real-time data, improving the robustness of the DOE calculation and helping mitigate the impact of forecast errors.

5.3. Network Models Used for DOE Calculation

When calculating DOE, it is important to adopt a network model to represent mutual phase couplings, as voltage imbalances are common in LV networks. The level of detail in network modeling varies across publications on DOE calculations. Some studies adopt a balanced (positive sequence) model, where network impedances are represented as a 1×1 impedance matrix. Others employ a symmetrical parameters-based approach, incorporating positive, negative, and zero sequence values to construct a 3×3 phase-coordinate impedance matrix. A more detailed representation includes a three-phase explicit-neutral model, forming a 4×4 impedance matrix derived from modified Carson's equations. This can be further simplified by using a Kron reduction, which eliminates the neutral node to obtain a 3×3 phase impedance matrix, reducing computational complexity while preserving the essential characteristics of the network [27]. However, this simplification may introduce approximation errors, particularly in networks with significant neutral currents or unbalanced loads, where an explicit-neutral representation remains more accurate. Hence it is important to utilize accurate network models that can closely represent the underlying electrical characteristics to ensure reliable DOE calculations, particularly in unbalanced networks where neutral currents play a significant role.

5.4. Uncertainties Associated with DOE Calculation

Most of the literature, such as [22,23,46], considers DSO to have full network visibility as well as perfectly accurate predicted demand and generation for each DOE calculation interval. However, this level of accuracy and visibility is rarely achievable. Authors have explored different ways of forecasting the required data for DOE calculations, such as statistical models, machine learning approaches [41], and real-time data collection using smart meters. Despite these efforts, the high intermittency of renewable energy sources,

combined with the inherent unpredictability of consumer demand patterns, makes it challenging to forecast these values with high precision. Uncertainties caused by stochastic DER generation, fluctuating consumer demand, as well as incorrect network model parameters, can cause the calculated DOEs to exceed the network constraints and cause potential voltage and thermal violations.

To address the uncertainty of forecasts for DOE calculation, Ref. [22] proposed a chance-constrained ACOPF, which ensures the constraints are satisfied within a certain probability. Their simulation results demonstrate that even with forecast errors, the calculated operating envelopes remain within voltage constraints. Ref. [47] explored how Polynomial Chaos Expansion (PCE)-based OPF can be extended to calculate day-ahead DOEs considering phase unbalance.

In DOE calculation It is often assumed if consumers follow their allocated operating envelopes, there will not be any network violations, which is true for balanced networks as there are no mutual phase couplings. But this is not the case for LV networks where there are significant voltage imbalances. A sensitivity-based analysis was conducted [20] to investigate how phase voltages respond to variations in customer demand. The findings revealed that voltage violations can happen even when consumers adhere to their allocated dynamic limits. The analysis, conducted on a 2-bus system, showed that increasing a customer's active demand—either by reducing exports or increasing imports—can help mitigate over-voltage issues in the corresponding phase. However, this adjustment may lead to over-voltage problems in the subsequent phase. Similarly, increasing a customer's reactive demand can alleviate over-voltage issues in the corresponding phase but may cause over-voltage problems in the preceding phase.

To address the challenges posed by forecasting errors and inaccurate network models; Ref. [48] developed a linear Unbalanced Three-Phase Optimal Power Flow (UTOPF)-based technique for calculating DOEs, considering both single (load forecasting errors) and combined uncertainties (load forecasting errors and inaccurate impedance values) constrained by norm balls. While applying a linear approach to the UTOPF introduces some inherent errors, it greatly improves computational tractability, making it more feasible for large-scale applications.

Ref. [49] proposed a method to calculate robust DOEs by first identifying a maximal hyperrectangle within the feasible region (FR), which involves solving a convex optimization problem to fit the largest hyper ellipsoid within the FR, then expanding this hyperrectangle by using Motzkin Transposition Theorem (MTT) to better utilize the network's capacity while maintaining proportional fairness in allocating DOEs among customers. This method explicitly incorporates optimization of controllable reactive power. It also leverages sensitivity analysis and constraint reduction to effectively manage large-scale networks, utilizing a linearized Unbalanced Three-Phase Optimal Power Flow (UTOPF) for improved performance. In [50], the authors address the sub-optimality of the solution proposed in [49] by redefining the Feasible Region (FR) as a super ellipsoid-based FR instead of a hyper ellipsoid. This new approach provides a near-optimal solution that is closer to global optimality. However, this improvement comes at the cost of increased computational time due to the more complex geometric representation and optimization process involved in handling the super ellipsoid. The authors in [51] introduced a calculation method for DOE under uncertainties that involves identifying worst-case active and reactive power utilization scenarios by analyzing phase voltage sensitivities to customer demands, which are computed through perturbations in demand and unbalanced three-phase flow (UTPF) simulations. To make the non-convex optimization tractable, sensitivity-based filtering and scenario reduction techniques are employed, reducing the number of scenarios by merging redundant ones and focusing on critical voltage constraints. Fairness is incor-

porated through various allocation strategies, such as maximum efficiency, proportional fairness, and α -fairness, which balance between efficiency and fairness while respecting operational constraints.

5.5. Data Privacy Issues

Cybersecurity is a critical concern for Dynamic Operating Envelopes (DOEs), as these systems rely on real-time or near real-time data exchange between aggregators, prosumers, and DSO. Potential cyber threats include unauthorized access to sensitive data, manipulation of data such as incorrect demand or generation profiles, denial-of-service (DoS) attacks disrupting communication, and malicious alterations of control systems affecting the scheduling of distributed energy resources (DERs) [52]. To ensure secure data transmission, strategies such as encryption and authentication are essential [53]. Additionally, intrusion detection systems (IDS) and anomaly detection can be deployed to monitor and identify unauthorized access [54]. Blockchain technology also offers a promising solution for decentralized data exchange, providing data integrity and preventing tampering [55] as proposed in [56]. Privacy concerns are particularly relevant in decentralized DOE computation, as real-time energy consumption data can reveal sensitive information about prosumers. To address these, data anonymization and aggregation techniques can prevent identification, while differential privacy can be applied to allow meaningful analysis without compromising individual privacy [57]. Furthermore, minimizing the data transfer between participating entities by using the Alternating Direction Method of Multipliers (ADMM) enhances privacy [58], as it allows decentralized computation with minimal data sharing while ensuring efficient optimization of DOE systems. By keeping data local and only exchanging necessary information, ADMM reduces the risks associated with unauthorized access and improves the overall security and privacy of the system.

6. Application of Dynamic Operating Envelopes

6.1. Local Energy Markets and DOE

In literature, Refs. [59–61] have thoroughly investigated local energy market mechanisms for active and reactive power trading within a Peer-to-Peer (P2P) network, which promotes a decentralized market as well as energy security. The advantages of deploying DOE-based Local Energy Market (LEM) among participating prosumers guarantee fairer access to network capacity, enhance local grid stability, and optimize the utilization of DERs as well as existing network assets. By incorporating DOEs into the LEM framework, prosumers can trade energy more efficiently while adhering to network constraints, thereby minimizing curtailment and maximizing self-consumption. Ref. [62] investigated how EVs and PV systems can be integrated into a LEM through a DOE-based framework. This integration allows the DSO to manage network operations by considering prosumers' expected energy imports/exports while resolving any network constraint violations using a DOE-based algorithm. A double auction-based P2P trading algorithm was introduced by [63] to facilitate energy transactions within DOEs, ensuring that network constraints are maintained while maximizing prosumer benefit. Ref. [56] proposed a game-theory-based approach where a Canonical Coalition Game (CCG) is formulated, allowing prosumers to collaborate and maximize their collective profits. The Shapley value is employed to ensure a fair distribution of profits among participants, allocating rewards based on each prosumer's contribution to the coalition. Ref. [64] further investigated the payoff allocation stability and fairness of the DOE integrated CCG-based P2P trading algorithm. Even though the proposed LEM algorithms improve financial gains for consumers, they rely on a centralized DSO to compute DOEs based on participants' proposed exports and imports,

introducing high computational and communication overhead, data privacy concerns, and dependency on a single authority, which may limit scalability and market fairness.

6.2. Demand Response Schemes and DOE

Demand Response (DR) is a consumer-focused strategy that has become increasingly popular with the development of smart grids and advanced metering infrastructure (AMI), which facilitate two-way communication between the grid and end-users through smart meters. DR programs generally allow customers to significantly impact grid operations by lowering or adjusting their electricity usage during peak times in response to time-based rates or other financial incentives. Ref. [65] utilized the DOE calculation method developed in [66] to determine DOE as a region instead of a set point. To achieve this, the maximum and minimum active and reactive power injections for each household were obtained. Latin Hypercube Sampling (LHS) was then used to generate sample set points, followed by the computation of the convex hull, which was aggregated using the Minkowski sum. These dynamic regions were subsequently communicated to the local controller for demand response (DR), making the resulting DR scheme model network aware and adaptive to changing network constraints.

7. Conclusions

DOE maximizes the DER integration to the grid within network constraints and can dynamically adjust export and import limits based on real-time network conditions and forecasted consumer demand and generation data. DOE enables the optimal utilization of the grid while mitigating network congestions, overutilization of network assets, and network constraint violations. This review has highlighted key methodologies utilized for DOE calculation, including optimization-based and data-driven approaches. Moreover, an investigation was carried out into the introduction of inherent errors resulting from the use of relaxations in OPF-based DOE calculations and the need for accurate network models. Furthermore, how DOE can be integrated within LEM and DR schemes was also studied. Despite significant progress, several challenges still need to be addressed. Uncertainties in network models, inaccurate load forecasts, and stochastic behavior of DER introduce complexities that require robust optimization and predictive control techniques, such as MPC. To this end, existing methods for addressing these uncertainties, such as robust DOE, were also examined. Additionally, developing approaches to mitigate these uncertainties without compromising computational speed or creating bottlenecks is crucial for minimizing errors in DOE calculations. Future research should focus on enhancing the scalability, interoperability, and decentralization of DOE frameworks while ensuring cybersecurity and regulatory compliance. Efforts should be made to improve the scalability of DOEs to effectively handle larger networks with higher DER penetration. Additionally, new DOE calculation methods are needed to address the stochastic nature of DER generation, and the volatile nature of customer demands without causing computational bottlenecks. Distributed DOE calculation methods will empower local stakeholders and foster greater energy autonomy. Moreover, striking the right balance between fairness and efficiency in DOE allocation will be crucial, as demonstrated by the conducted case study. Finally, ensuring the cybersecurity of DOE systems, particularly in decentralized settings, is essential to prevent potential vulnerabilities. These advancements will be critical to enhancing the effectiveness and broader adoption of DOE in modern power grids.

Author Contributions: Conceptualization, A.W., M.V., G.N. and P.C.; methodology, A.W.; software, A.W.; validation, A.W., M.V., G.N. and P.C.; formal analysis, A.W.; investigation, A.W.; resources, M.V., G.N. and P.C; data curation, A.W.; writing—original draft preparation, A.W.; writing—review

and editing, A.W., M.V., G.N. and P.C; visualization, A.W.; supervision, M.V., G.N. and P.C. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: No data were used for this research.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

DOE	Dynamic Operating Envelopes
DER	Distributed Energy Resources
OPF	Optimal Power Flow
PV	Photo-Voltaic
BESS	Battery Energy Storage Systems
EV	Electric Vehicles
DSO	Distribution Service Operators
OLTC	On-Load Tap Changer
DNSP	Distribution Network Service Providers
EDGE	Energy Demand and Generation Exchange
EVOLVE	Project Energy Value Optimization Leveraging Envelopes
ARENA	Australian Renewable Energy Agency
ACOPF	Alternating Current Optimal Power Flow
AEMO	Australian Energy Market Operator
SOE	Static Operating Envelopes
NEM	National Electricity Market
DSSE	Distribution System State Estimation
VUF	Voltage Unbalance Factor
ML	Machine Learning
NN	Neural Networks
SOCP	Second-Order Cone Programming
MPC	Model Predictive Control
PCE	Polynomial Chaos Expansion
UTOPF	Unbalanced Three-Phase Optimal Power Flow
FR	Feasible Region
MTT	Motzkin Transposition Theorem
UTPF	Unbalanced Three-Phase Power Flow
IDS	Intrusion Detection Systems
ADMM	Alternating Direction Method of Multipliers
LEM	Local Energy Markets
P2P	Peer-to-Peer
CCG	Canonical Coalition Game
DR	Demand Response
AMI	Advanced Metering Infrastructure
LHS	Latin Hypercube Sampling

References

1. DEIP Dynamic Operating Envelopes Workstream. Available online: <https://arena.gov.au/assets/2021/09/doe-workshop-summary.pdf> (accessed on 21 March 2024).
2. Haque, M.M.; Wolfs, P. A review of high PV penetrations in LV distribution networks: Present status, impacts and mitigation measures. *Renew. Sustain. Energy Rev.* **2016**, *62*, 1195–1208. [CrossRef]
3. Chathurangi, D.; Jayatunga, U.; Perera, S.; Agalgaonkar, A.P.; Siyambalapitiya, T. Comparative evaluation of solar PV hosting capacity enhancement using Volt-VAr and Volt-Watt control strategies. *Renew. Energy* **2021**, *177*, 1063–1075. [CrossRef]

4. Dynamic Standard for Small IES Connections. Available online: https://www.energex.com.au/__data/assets/pdf_file/0008/1072592/STNW3510-Dynamic-Standard-for-Small-IES-Connections.pdf (accessed on 5 June 2024).
5. Bozalakov, D.; Mnati, M.J.; Laveyne, J.; Desmet, J.; Vandevelde, L. Battery Storage Integration in Voltage Unbalance and Overvoltage Mitigation Control Strategies and Its Impact on the Power Quality. *Energies* **2019**, *12*, 1501. [CrossRef]
6. Alajrash, B.H.; Salem, M.; Swadi, M.; Senjyu, T.; Kamarol, M.; Motahhir, S. A comprehensive review of FACTS devices in modern power systems: Addressing power quality, optimal placement, and stability with renewable energy penetration. *Energy Rep.* **2024**, *11*, 5350–5371. [CrossRef]
7. On the Calculation and Use of Dynamic Operating Envelopes. Available online: <https://arena.gov.au/assets/2020/09/on-the-calculation-and-use-of-dynamic-operating-envelopes.pdf> (accessed on 27 June 2024).
8. Petrou, K.; Procopiou, A.T.; Gutierrez-Lagos, L.; Liu, M.Z.; Ochoa, L.F.; Langstaff, T.; Theunissen, J.M. Ensuring Distribution Network Integrity Using Dynamic Operating Limits for Prosumers. *IEEE Trans. Smart Grid* **2021**, *12*, 3877–3888. [CrossRef]
9. Liu, M.Z.; Ochoa, L.N.; Riaz, S.; Mancarella, P.; Ting, T.; San, J.; Theunissen, J. Grid and Market Services from the Edge: Using Operating Envelopes to Unlock Network-Aware Bottom-Up Flexibility. *IEEE Power Energy Mag.* **2021**, *19*, 52–62. [CrossRef]
10. Project EDGE (Energy Demand and Generation Exchange)—Australian Renewable Energy Agency (ARENA). Available online: <https://arena.gov.au/projects/project-edge-energy-demand-and-generation-exchange/> (accessed on 28 January 2025).
11. “Evolve—Battery Storage and Grid Integration Program”. Battery Storage and Grid Integration Program—An initiative of the Australian National University. Available online: <https://bsgip.com/research/evolve/> (accessed on 2 March 2025).
12. Home-Australian Renewable Energy Agency. Available online: <https://arena.gov.au/> (accessed on 2 March 2025).
13. Open Energy Networks Project—Energy Networks Australia Position Paper. Available online: <https://www.energynetworks.com.au/resources/reports/2020-reports-and-publications/open-energy-networks-project-energy-networks-australia-position-paper/> (accessed on 7 September 2024).
14. Energex-About Dynamic Connections. Available online: <https://www.energex.com.au/our-services/connections/residential-and-commercial-connections/solar-connections-and-other-technologies/dynamic-connections-for-energy-exports/about-dynamic-connections> (accessed on 17 May 2024).
15. Accelerating the Implementation of Operating Envelopes Across Australia—Milestone 4: Final Report. Available online: <https://www.csiro.au/-/media/EF/Files/GPST-Roadmap/Stage3-Final/Topic-8Distributed-Energy-Resources-DErs.pdf> (accessed on 4 November 2024).
16. NEM Data Dashboard. Available online: <https://aemo.com.au/energy-systems/electricity/national-electricity-market-nem/data-nem/data-dashboard-nem> (accessed on 2 March 2025).
17. Assessing the Benefits of Using Operating Envelopes to Orchestrate DERs Across Australia. Available online: <https://www.csiro.au/-/media/EF/Files/GPST-Roadmap/Final-Reports/Topic-8-GPST-Stage-2.pdf> (accessed on 5 August 2024).
18. Alam, M.R.; Nguyen, P.T.H.; Naranpanawe, L.; Saha, T.K.; Lankeshwara, G. Allocation of Dynamic Operating Envelopes in Distribution Networks: Technical and Equitable Perspectives. *IEEE Trans. Sustain. Energy* **2023**, *15*, 173–186. [CrossRef]
19. Milford, T.; Krause, O. Managing DER in Distribution Networks Using State Estimation & Dynamic Operating Envelopes. In Proceedings of the 2021 IEEE PES Innovative Smart Grid Technologies—Asia (ISGT Asia), Brisbane, Australia, 5–8 December 2021.
20. Liu, B.; Braslavsky, J.H. Sensitivity and Robustness Issues of Operating Envelopes in Unbalanced Distribution Networks. *IEEE Access* **2022**, *10*, 92789–92798. [CrossRef]
21. Moring, H.; Mathieu, J.L. Inexactness of Second Order Cone Relaxations for Calculating Operating Envelopes. In Proceedings of the 2023 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), Glasgow, UK, 31 October–3 November 2023.
22. Yi, Y.; Verbič, G. Fair operating envelopes under uncertainty using chance constrained optimal power flow. *Electr. Power Syst. Res.* **2022**, *213*, 108465. [CrossRef]
23. Lankeshwara, G. A Real-Time Control Approach to Maximise the Utilisation of Rooftop PV Using Dynamic Export Limits. In Proceedings of the 2021 IEEE PES Innovative Smart Grid Technologies—Asia (ISGT Asia), Brisbane, Australia, 5–8 December 2021.
24. Decision Regulatory Impact Statement 2017. Available online: <https://cabinet.qld.gov.au/documents/2017/Sep/Voltage/Attachments/DecisionRIS.PDF> (accessed on 2 August 2024).
25. A Gentle Introduction to Power Flow. Available online: <https://invenia.github.io/blog/2020/12/04/pf-intro/> (accessed on 2 March 2025).
26. Farivar, M.; Low, S.H. Branch Flow Model: Relaxations and Convexification—Part I. *IEEE Trans. Power Syst.* **2013**, *28*, 2554–2564. [CrossRef]
27. Antić, T.; Geth, F.; Capuder, T. The Importance of Technical Distribution Network Limits in Dynamic Operating Envelopes. In Proceedings of the 2023 IEEE Belgrade PowerTech, Belgrade, Serbia, 25–29 June 2023.
28. Bassi, V.; Ochoa, L.F.; Alpcan, T.; Leckie, C. Electrical Model-Free Voltage Calculations Using Neural Networks and Smart Meter Data. *IEEE Trans. Smart Grid* **2022**, *14*, 3271–3282. [CrossRef]

29. Final Report: Model-Free Operating Envelopes. 2023. Available online: https://c4net.com.au/wp-content/uploads/2023/09/UoM-C4NET-Model-Free_OEs-FinalReport_v02.pdf (accessed on 2 May 2024).
30. Kumarawadu, A.; Azim, M.I.; Khorasany, M.; Razzaghi, R.; Heidari, R. Smart meter data-driven dynamic operating envelopes for DERs. *Appl. Energy* **2025**, *384*, 125469.
31. Carpentier, J.L. Optimal Power Flows: Uses, Methods and Developments. *IFAC Proc. Vol.* **1985**, *18*, 11–21. [\[CrossRef\]](#)
32. Xia, X.; Elaiw, A.M. Optimal dynamic economic dispatch of generation: A review. *Electr. Power Syst. Res.* **2010**, *80*, 975–986. [\[CrossRef\]](#)
33. Liu, M.Z.; Ochoa, L.F.; Wong, P.K.C.; Theunissen, J. Using OPF-Based Operating Envelopes to Facilitate Residential DER Services. *IEEE Trans. Smart Grid* **2022**, *13*, 4494–4504. [\[CrossRef\]](#)
34. Lavaei, J.; Low, S.H. Zero Duality Gap in Optimal Power Flow Problem. *IEEE Trans. Power Syst.* **2011**, *27*, 92–107.
35. Low, S.H. Convex Relaxation of Optimal Power Flow—Part I: Formulations and Equivalence. *IEEE Trans. Control Netw. Syst.* **2014**, *1*, 15–27. [\[CrossRef\]](#)
36. Wang, L.; Yan, R.; Bai, F.; Saha, T.; Wang, K. A Distributed Inter-Phase Coordination Algorithm for Voltage Control with Unbalanced PV Integration in LV Systems. *IEEE Trans. Sustain. Energy* **2020**, *11*, 2687–2697. [\[CrossRef\]](#)
37. Australian Photovoltaic Institute. Available online: <https://pv-map.apvi.org.au/> (accessed on 17 December 2024).
38. Flexible Exports. Available online: <https://www.ausnetservices.com.au/flexible-exports> (accessed on 17 May 2024).
39. Review of Dynamic Operating Envelope Adoption by DNSPs. Available online: <https://arena.gov.au/assets/2022/07/review-of-dynamic-operating-envelopes-from-dnsp.pdf> (accessed on 15 November 2024).
40. Project EDGE—Determining the Impact of Update Frequency on Operating Envelope Efficacy. Available online: <https://aemo.com.au/initiatives/major-programs/nem-distributed-energy-resources-der-program/der-demonstrations/project-edge/project-edge-reports/dynamic-operating-envelopes> (accessed on 25 September 2024).
41. Lanas, F.; Liu, M.; Ochoa, L.F. Understanding the Role of Forecast Errors on the Effectiveness of Operating Envelopes. In Proceedings of the 2024 IEEE Power & Energy Society General Meeting (PESGM), Seattle, WA, USA, 21–25 July 2024.
42. Hong, T.; Fan, S. Probabilistic electric load forecasting: A tutorial review. *Int. J. Forecast.* **2016**, *32*, 914–938. [\[CrossRef\]](#)
43. Hooshmand Pakdel, G.; He, Y.; Chen, X. Predicting customer demand with deep learning: An LSTM-based approach incorporating customer information. *Int. J. Prod. Res.* **2015**, 1–13.
44. Rodríguez, F.; Fleetwood, A.; Galarza, A.; Fontán, L. Predicting solar energy generation through artificial neural networks using weather forecasts for microgrid control. *Renew. Energy* **2018**, *126*, 855–864. [\[CrossRef\]](#)
45. Cavus, M.; Allahham, A. Enhanced Microgrid Control through Genetic Predictive Control: Integrating Genetic Algorithms with Model Predictive Control for Improved Non-Linearity and Non-Convexity Handling. *Energies* **2024**, *17*, 4458. [\[CrossRef\]](#)
46. Petrou, K.; Liu, M.Z.; Procopiou, A.T.; Ochoa, L.F.; Theunissen, J.; Harding, J. Operating Envelopes for Prosumers in LV Networks: A Weighted Proportional Fairness Approach. In Proceedings of the 2020 IEEE PES Innovative Smart Grid Technologies Europe (ISGT-Europe), The Hague, The Netherlands, 26–28 October 2020.
47. Koirala, A.; Geth, F.; Van Acker, T. Day-ahead dynamic operating envelopes using stochastic unbalanced optimal power flow. *Sustainable Energy, Grids and Networks* **2024**, *40*, 101528. [\[CrossRef\]](#)
48. Liu, B.; Braslavsky, J.H.; Mahdavi, N. Linear OPF-Based Robust Dynamic Operating Envelopes with Uncertainties in Unbalanced Distribution Networks. *J. Mod. Power Syst. Clean Energy* **2023**, *12*, 1320–1326.
49. Liu, B.; Braslavsky, J.H. Robust Dynamic Operating Envelopes for DER Integration in Unbalanced Distribution Networks. *IEEE Trans. Power Syst.* **2023**, *39*, 3921–3936. [\[CrossRef\]](#)
50. Liu, B.; Braslavsky, J.H. Robust Dynamic Operating Envelopes via Superellipsoid-Based Convex Optimisation in Unbalanced Distribution Networks. *IEEE Trans. Power Syst.* **2024**, *39*, 4775–4778.
51. Liu, B.; Braslavsky, J.H. Computation of robust dynamic operating envelopes based on non-convex OPF for unbalanced distribution networks. *CSEE J. Power Energy Syst.* **2024**; early access.
52. Diaba, S.Y.; Shafie-Khah, M.; Elmusrati, M. Cyber-physical attack and the future energy systems: A review. *Energy Rep.* **2024**, *12*, 2914–2932. [\[CrossRef\]](#)
53. Implementing Certificates, TLS, HTTPS and Opportunistic TLS. Available online: [https://www.cyber.gov.au/sites/default/files/2023-03/PROTECT%20-%20Implementing%20Certificates,%20TLS,%20HTTPS%20and%20Opportunistic%20TLS%20\(October%202021\).pdf](https://www.cyber.gov.au/sites/default/files/2023-03/PROTECT%20-%20Implementing%20Certificates,%20TLS,%20HTTPS%20and%20Opportunistic%20TLS%20(October%202021).pdf) (accessed on 20 March 2025).
54. Anomaly Detection in Intrusion Detection Systems. Available online: <https://www.intechopen.com/chapters/88343> (accessed on 20 March 2025).
55. What Is Blockchain? Available online: <https://www.ibm.com/think/topics/blockchain#:~:text=Blockchain%20is%20a%20shared,%20immutable,patents,%20copyrights,%20branding> (accessed on 20 March 2025).
56. Azim, M.I.; Lankeshwara, G.; Tushar, W.; Sharma, R.; Alam, M.R.; Saha, T.K.; Khorasany, M.; Razzaghi, R. Dynamic Operating Envelope-Enabled P2P Trading to Maximize Financial Returns of Prosumers. *IEEE Trans. Smart Grid* **2024**, *15*, 1978–1990. [\[CrossRef\]](#)

57. Dyda, A.; Purcell, M.; Curtis, S.; Field, E.; Pillai, P.; Ricardo, K.; Weng, H.; Moore, J.C.; Hewett, M.; Williams, G.; et al. Differential privacy for public health data: An innovative tool to optimize information sharing while protecting data confidentiality. *Patterns* **2021**, *2*, 100366.
58. Lankeshwara, G.; Sharma, R. Dynamic Operating Envelopes-enabled Demand Response in Low-voltage Residential Networks. In Proceedings of the 2022 IEEE PES 14th Asia-Pacific Power and Energy Engineering Conference (APPEEC), Melbourne, Australia, 20–23 November 2022.
59. Dudjak, V.; Neves, D.; Alskaf, T.; Khadem, S.; Pena-Bello, A.; Saggese, P.; Bowler, B.; Andoni, M.; Bertolini, M.; Zhou, Y.; et al. Impact of local energy markets integration in power systems layer: A comprehensive review. *Appl. Energy* **2021**, *301*, 117434. [\[CrossRef\]](#)
60. Capper, T.; Gorbacheva, A.; Mustafa, M.A.; Bahloul, M.; Schwidtal, J.M.; Chitchyan, R.; Andoni, M.; Robu, V.; Montakhabi, M.; Scott, I.J.; et al. Peer-to-peer, community self-consumption, and transactive energy: A systematic literature review of local energy market models. *Renew. Sustain. Energy Rev.* **2022**, *162*, 112403. [\[CrossRef\]](#)
61. Jogunola, O.; Ajagun, A.S.; Tushar, W.; Olatunji, F.O.; Yuen, C.; Morley, C.; Adebisi, B.; Shongwe, T. Peer-to-Peer Local Energy Market: Opportunities, Barriers, Security, and Implementation Options. *IEEE Access* **2024**, *12*, 37873–37890. [\[CrossRef\]](#)
62. Hoque, M.M.; Khorasany, M.; Azim, M.I.; Razzaghi, R.; Jalili, M. Dynamic Operating Envelope-Based Local Energy Market for Prosumers with Electric Vehicles. *IEEE Trans. Smart Grid* **2023**, *15*, 1712–1724. [\[CrossRef\]](#)
63. Hoque, M.M.; Khorasany, M.; Azim, M.I.; Razzaghi, R.; Jalili, M. A framework for prosumer-centric peer-to-peer energy trading using network-secure export–import limits. *Appl. Energy* **2024**, *361*, 122906. [\[CrossRef\]](#)
64. Azim, M.I.; Hoque, M.M.; Khorasany, M.; Razzaghi, R.; Jalili, M.; Hill, D.J. Dynamic Operating Envelope-Integrated Cooperative Energy Trading: A Fair Allocation Approach. *IEEE Trans. Power Syst.* **2024**, *early access*.
65. Lankeshwara, G.; Sharma, R.; Alam, M.R.; Yan, R.; Saha, T.K. Development and validation of a dynamic operating envelopes-enabled demand response scheme in low-voltage distribution networks. *Appl. Energy* **2025**, *382*, 125150. [\[CrossRef\]](#)
66. Lankeshwara, G.; Sharma, R.; Yan, R.; Saha, T.K.; Milanović, J.V. Time-Varying Operating Regions of End-Users and Feeders in Low-Voltage Distribution Networks. *IEEE Trans. Power Syst.* **2023**, *39*, 4600–4611. [\[CrossRef\]](#)

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